

# Lattice QCD Data Summary Report

## Chiral Condensates and Susceptibilities Across Finite Temperature

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### Abstract

This report provides numerical measurement data tables and temperature-dependent figures for: (1) dimensionless renormalized subtracted chiral condensates  $\Delta_{l,s}$ , (2) temporal scaling series at  $\beta = 4.17$ , and (3) light-quark chiral susceptibilities  $\chi_{\text{scaled}}$  [MeV<sup>2</sup>] computed with unbiased quadratic estimators across finite temperatures.

## 1 Definitions and Observables

### 1.1 Chiral Condensates

The single-flavor bare chiral condensate on a lattice volume  $V = N_s^3 \times N_t$  is defined as

$$\langle \bar{\psi}\psi \rangle_q = \frac{1}{V} \langle \text{Tr} (D_q^{-1}) \rangle, \quad q \in \{l, s\} \quad (1)$$

evaluated via stochastic  $Z_2$  volume source vectors.

The renormalized subtracted chiral condensate  $\Delta_{l,s}$  eliminates additive ultraviolet divergences and residual chiral symmetry breaking:

$$\Delta_{l,s}(T) = \frac{1}{Z_m(\beta)} \left[ \langle \bar{\psi}\psi \rangle_l - \frac{m_l + m_{\text{res}}}{m_s + m_{\text{res}}} \langle \bar{\psi}\psi \rangle_s \right] \quad (2)$$

where  $m_{\text{res}}(\beta)$  is the residual quark mass, and  $Z_m(\beta)$  is the scalar mass renormalization factor. In lattice QCD thermodynamics,  $\Delta_{l,s}$  directly serves as the dimensionless order parameter for chiral symmetry restoration.

### 1.2 Chiral Susceptibility

The light-quark disconnected chiral susceptibility  $\chi$  is defined as

$$\chi = \frac{V}{T} \left[ \left\langle \left( \frac{1}{V} \text{Tr}(D_l^{-1}) \right)^2 \right\rangle - \left\langle \frac{1}{V} \text{Tr}(D_l^{-1}) \right\rangle^2 \right] \quad (3)$$

Using  $k$  stochastic volume noise vectors  $O_{l,i}$  on each gauge configuration without vector-level RM subtraction, the noise bias is removed by the unbiased quadratic cross-product estimator:

$$\bar{O} = \frac{1}{k} \sum_{i=1}^k O_{l,i}, \quad \overline{O^2} = \frac{1}{k(k-1)} \sum_{i \neq j} O_{l,i} O_{l,j} = \frac{1}{k(k-1)} \left[ \left( \sum_{i=1}^k O_{l,i} \right)^2 - \sum_{i=1}^k O_{l,i}^2 \right] \quad (4)$$

The ensemble susceptibility and its Jackknife statistical error are given by

$$a_r = \text{JK}(\bar{O})_r, \quad b_r = \text{JK}(\overline{O^2})_r, \quad \chi_r = b_r - a_r^2 \quad (5)$$

$$\bar{\chi} = \frac{1}{N} \sum_{r=1}^N \chi_r, \quad \sigma_\chi = \sqrt{\frac{N-1}{N} \sum_{r=1}^N (\chi_r - \bar{\chi})^2} \quad (6)$$

The continuous thermal scaled susceptibility with physical dimensions of [MeV<sup>2</sup>] is

$$\chi_{\text{scaled}} = (16T)^2 a^2 \chi = N_s^3 \times N_t^3 \times T^2 \times \chi \quad [\text{MeV}^2] \quad (7)$$

## 2 Chiral Condensate Data and Temperature Dependence

### 2.1 Numerical Data Tables

Table 1 presents the dimensionless renormalized subtracted chiral condensate  $\Delta_{l,s}$  across the 8 standard temperature points on  $48^3 \times 16$  lattices. Table 2 summarizes the results for the finite temporal extent scaling ensembles at  $\beta = 4.17$ .

Table 1: Dimensionless renormalized subtracted chiral condensate  $\Delta_{l,s}$  across 8 temperature points on standard  $48^3 \times 16$  lattices.

| $\beta$     | $T$ [MeV]     | Geometry ( $N_s^3 \times N_t$ )    | $N_{\text{cfg}}$ | $\Delta_{l,s}$                                         |
|-------------|---------------|------------------------------------|------------------|--------------------------------------------------------|
| 4.13        | 138.23        | $48^3 \times 16$                   | 2000             | $(1.0608 \pm 0.0067) \times 10^{-3}$                   |
| 4.15        | 145.75        | $48^3 \times 16$                   | 2024             | $(7.7109 \pm 0.0675) \times 10^{-4}$                   |
| 4.17        | 153.31        | $48^3 \times 16$                   | 2008             | $(5.0728 \pm 0.0662) \times 10^{-4}$                   |
| <b>4.18</b> | <b>157.03</b> | <b><math>48^3 \times 16</math></b> | <b>2029</b>      | <b><math>(4.1649 \pm 0.0659) \times 10^{-4}</math></b> |
| 4.20        | 164.55        | $48^3 \times 16$                   | 2072             | $(2.3896 \pm 0.0561) \times 10^{-4}$                   |
| 4.23        | 176.43        | $48^3 \times 16$                   | 2012             | $(9.2886 \pm 0.3810) \times 10^{-5}$                   |
| 4.30        | 202.52        | $48^3 \times 16$                   | 2052             | $(8.2043 \pm 1.3300) \times 10^{-6}$                   |
| 4.405       | 241.60        | $48^3 \times 16$                   | 4859             | $(3.4370 \pm 1.0350) \times 10^{-7}$                   |

Table 2: Dimensionless renormalized subtracted chiral condensate  $\Delta_{l,s}$  for finite temporal scaling ensembles at  $\beta = 4.17$  ( $m_l = 0.0020, m_s = 0.0400$ ).

| Ensemble          | Geometry         | $T$ [MeV] | $N_{\text{cfg}}$ | $\Delta_{l,s}$                        |
|-------------------|------------------|-----------|------------------|---------------------------------------|
| L48T18 (stream 1) | $48^3 \times 18$ | 136.28    | 98               | $(8.2980 \pm 0.4422) \times 10^{-4}$  |
| L48T18 (stream 2) | $48^3 \times 18$ | 136.28    | 76               | $(7.9475 \pm 4.6593) \times 10^{-5}$  |
| L40T16            | $40^3 \times 16$ | 153.31    | 879              | $(3.3260 \pm 0.7358) \times 10^{-5}$  |
| L32T14            | $32^3 \times 14$ | 175.21    | 291              | $(-1.1468 \pm 1.1468) \times 10^{-8}$ |
| L32T12            | $32^3 \times 12$ | 204.41    | 108              | $(-2.0497 \pm 1.8947) \times 10^{-5}$ |

## 2.2 Plots vs Temperature

### 2.2.1 Standard $48^3 \times 16$ Ensembles

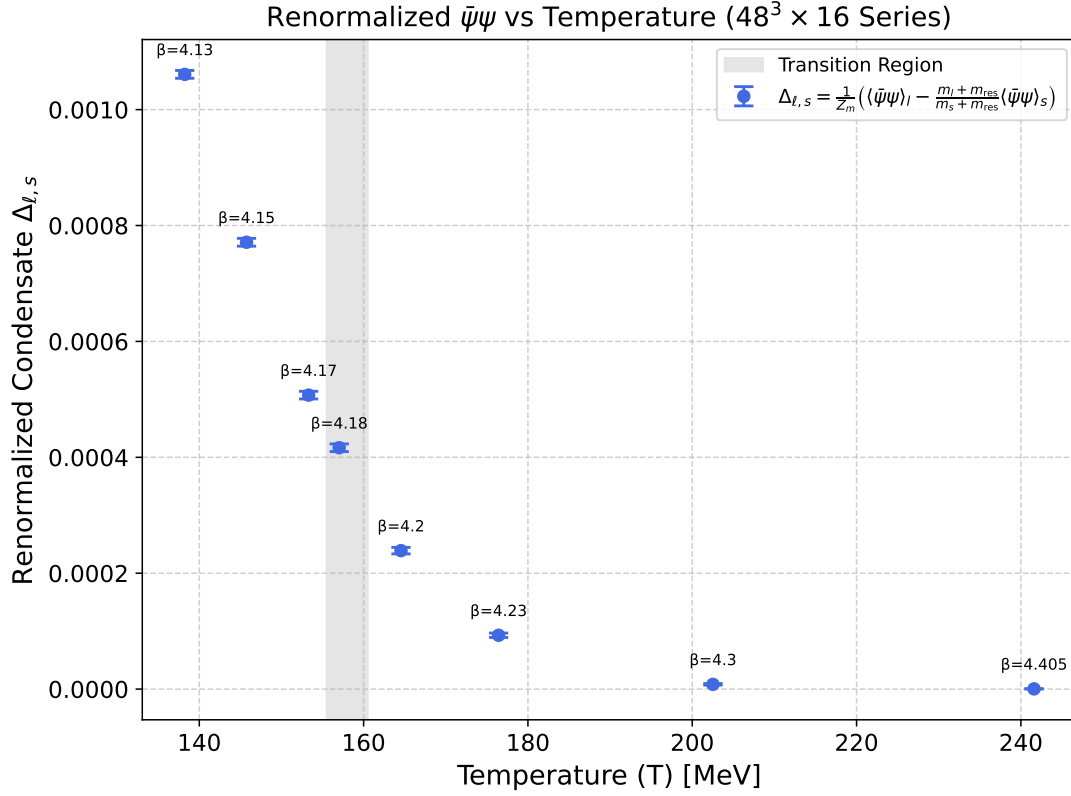


Figure 1: Dimensionless renormalized subtracted chiral condensate  $\Delta_{l,s}$  as a function of temperature  $T$  [MeV] for standard  $48^3 \times 16$  lattices. The shaded band marks the pseudo-critical transition window  $T \in [155.5, 160.5]$  MeV.

### 2.2.2 Finite Temporal Scaling Ensembles ( $\beta = 4.17$ )

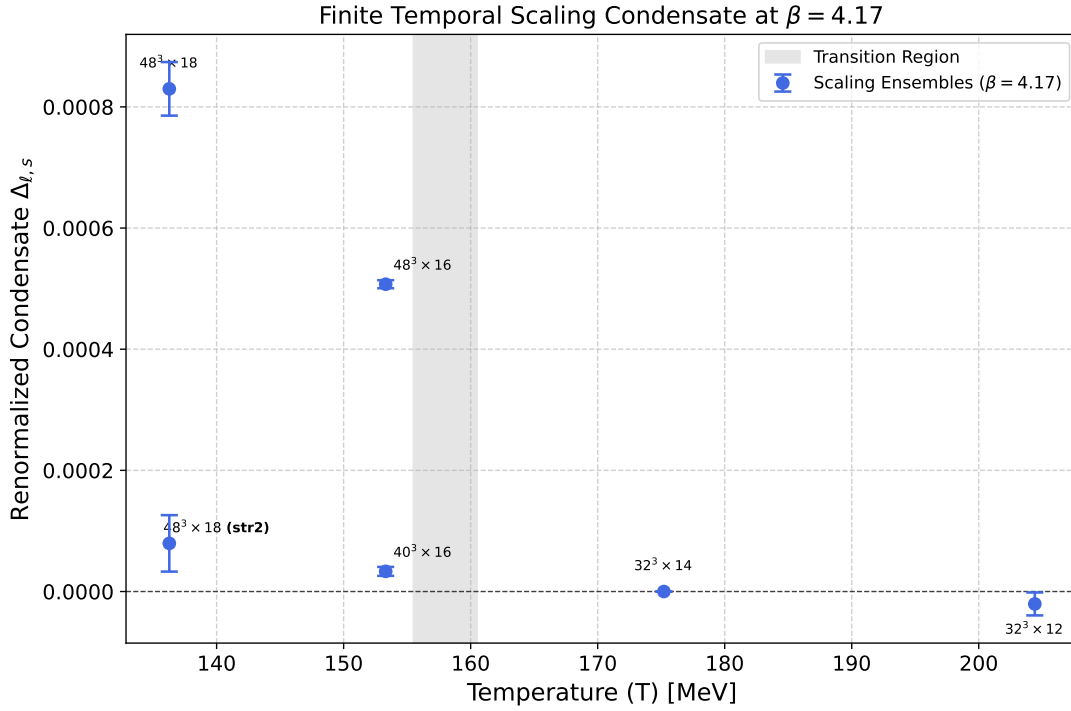


Figure 2: Dimensionless renormalized subtracted chiral condensate  $\Delta_{l,s}$  across temporal geometries  $N_t = 12, 14, 16, 18$  ( $T = 153.31 \times 16/N_t$  [MeV]) at fixed  $\beta = 4.17$ .

## 3 Chiral Susceptibility Data and Temperature Dependence

### 3.1 Numerical Data Tables

Table 3 presents the continuous scaled chiral susceptibility  $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$  across the 8 standard temperature points ( $48^3 \times 16$ ). Table 4 lists the chiral susceptibility values computed for the finite temporal scaling ensembles at  $\beta = 4.17$ .

Table 3: Continuous scaled chiral susceptibility  $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$  on standard  $48^3 \times 16$  lattices (unbiased quadratic estimator without RM subtraction).

| $\beta$     | $T$ [MeV]     | Geometry ( $N_s^3 \times N_t$ )    | $N_{\text{cfg}}$ | $\chi_{\text{scaled}} [(16T)^2 a^2 \chi]$ [MeV <sup>2</sup> ] |
|-------------|---------------|------------------------------------|------------------|---------------------------------------------------------------|
| 4.13        | 138.23        | $48^3 \times 16$                   | 2000             | $(6.6216 \pm 0.2307) \times 10^5$                             |
| 4.15        | 145.75        | $48^3 \times 16$                   | 2024             | $(7.8002 \pm 0.2673) \times 10^5$                             |
| 4.17        | 153.31        | $48^3 \times 16$                   | 2008             | $(8.5828 \pm 0.2577) \times 10^5$                             |
| <b>4.18</b> | <b>157.03</b> | <b><math>48^3 \times 16</math></b> | <b>2029</b>      | <b><math>(9.1495 \pm 0.3229) \times 10^5</math></b>           |
| 4.20        | 164.55        | $48^3 \times 16$                   | 2072             | $(7.6595 \pm 0.2769) \times 10^5$                             |
| 4.23        | 176.43        | $48^3 \times 16$                   | 2012             | $(4.0906 \pm 0.2315) \times 10^5$                             |
| 4.30        | 202.52        | $48^3 \times 16$                   | 2052             | $(7.1921 \pm 1.5027) \times 10^4$                             |
| 4.405       | 241.60        | $48^3 \times 16$                   | 4856             | $(8.6496 \pm 1.6936) \times 10^1$                             |

Table 4: Continuous scaled chiral susceptibility  $\chi_{\text{scaled}}$  for finite temporal scaling ensembles at  $\beta = 4.17$  ( $m_l = 0.0020, m_s = 0.0400$ ).

| Ensemble          | Geometry         | $T$ [MeV] | $N_{\text{cfg}}$ | $\chi_{\text{scaled}} [(16T)^2 a^2 \chi]$ [MeV <sup>2</sup> ] |
|-------------------|------------------|-----------|------------------|---------------------------------------------------------------|
| L48T18 (stream 1) | $48^3 \times 18$ | 136.28    | 79               | $(3.1489 \pm 0.4487) \times 10^5$                             |
| L48T18 (stream 2) | $48^3 \times 18$ | 136.28    | 72               | $(3.0177 \pm 0.5562) \times 10^5$                             |
| L40T16            | $40^3 \times 16$ | 153.31    | 17               | $(2.8070 \pm 0.8693) \times 10^5$                             |
| L32T14            | $32^3 \times 14$ | 175.21    | 282              | $(4.2497 \pm 0.6065) \times 10^5$                             |
| L32T12            | $32^3 \times 12$ | 204.41    | 79               | $(8.9135 \pm 4.9052) \times 10^4$                             |

## 3.2 Plots vs Temperature

### 3.2.1 Standard $48^3 \times 16$ Lattices

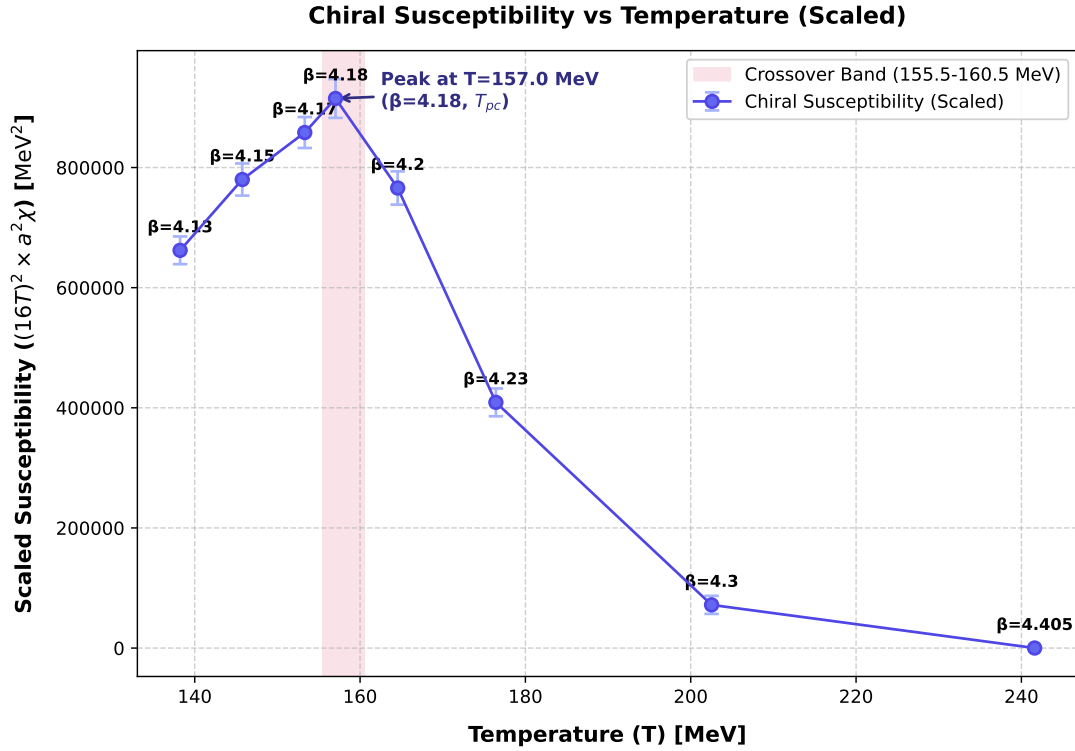


Figure 3: Continuous scaled chiral susceptibility  $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$  [MeV<sup>2</sup>] as a function of temperature  $T$  [MeV] for standard  $48^3 \times 16$  lattices. The peak marks the pseudo-critical transition temperature  $T_{pc} = 157.03$  MeV ( $\beta = 4.18$ ). The shaded band denotes the crossover window  $T \in [155.5, 160.5]$  MeV.

### 3.2.2 Finite Temporal Scaling Ensembles ( $\beta = 4.17$ )

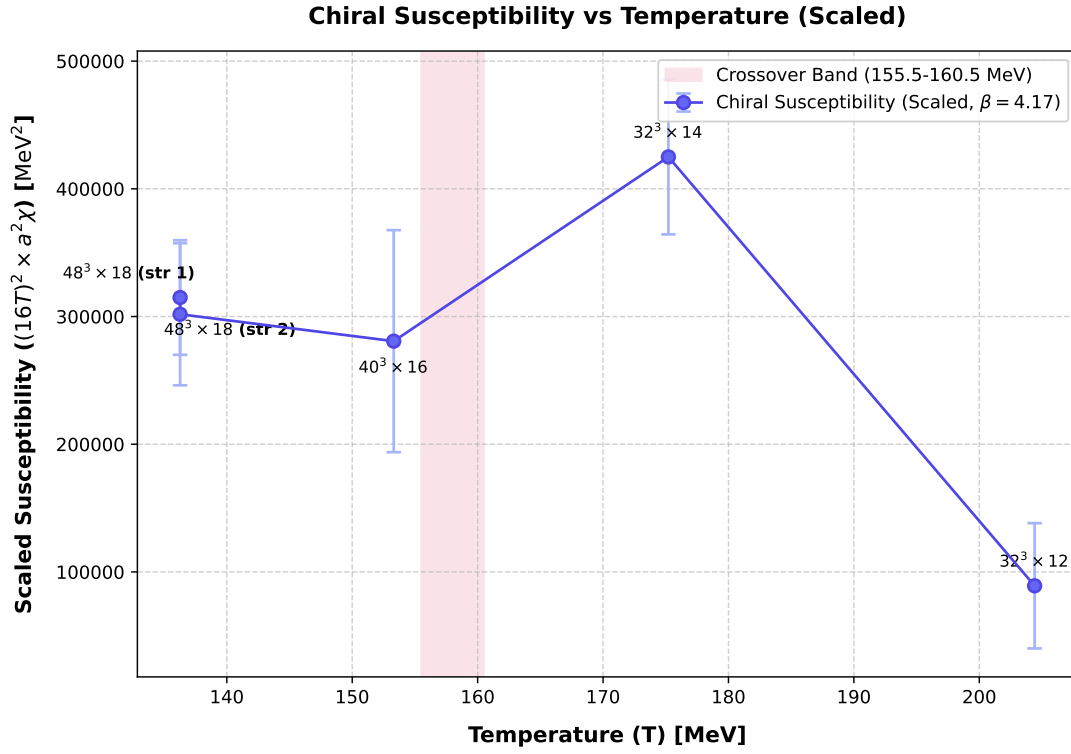


Figure 4: Continuous scaled chiral susceptibility  $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$  [MeV<sup>2</sup>] as a function of temperature  $T$  [MeV] for finite temporal scaling ensembles at  $\beta = 4.17$  ( $m_l = 0.0020, m_s = 0.0400$ ).

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